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Fault diagnosis of rolling element bearing using continuous wavelet transform and K- nearest neighbour

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ABSTRACT

Rotating machines are a common class for machineries in industries and one of the root cause of failure of these machines are faults in rolling element bearing (REB). Therefore, to prevent machinery performance degradation and malfunctioning, an effective bearing fault diagnosis technique is vital. So, the use of different machine learning techniques proposes a new method of diagnosing REB defects. Machine learning (ML) method namely K-nearest neighbour (KNN) is used to classify different types of REB faults. In this article, vibration signals are acquired from a customized test rig for three conditions of deep groove ball bearings: Normal (N), defect on Inner Race (IR) and defect on Outer Race (OR) at different loads and maximum speed. Continuous Wavelet Transform (CWT) has been applied to the acquired signal and Seventeen Time Frequency Domain (TFD) statistical features were extracted from CWT coefficients. KNN classifier is trained using these TFD. The classification accuracy on test data i.e., 83.3% reveals that KNN can be effectively used to classify different conditions of REB.
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1. Introduction

Rolling Element Bearings (REB) have been widely used in various industries, ranging from simple induction motor to complex manufacturing facilities. Thus, it is essential to implement an effective REB maintenance strategy to minimize the impact of failures due to malfunctioning. In the past, authors [1] compared various maintenance strategies and confirmed that condition monitoring had minimal extended damages and downtime. Various condition monitoring techniques are used in industries such as lubricant analysis, acoustic emission, vibration analysis, ultrasonic testing, and infrared thermography. For rotary machines, vibration analysis is widely used because of its ability to provide information on irregularities on the internal structure of the bearing [2]. In vibration analysis, time, frequency, and time–frequency domains were widely used. Among them, time frequency is superior as it allows the investigation of transient features such as shocks and impacts. Some of the time–frequency domain techniques include Short-time Fourier Transform (STFT), Wigner-Ville distribution (WVD), Choi-William Distribution (CWD) and the Wavelet Transform (WT). Unlike WT, other analysis techniques fail to reveal aspects of data such as trends, breakdown points and discontinuities in higher derivatives. WT can be further classified as CWT, Discrete Wavelet Transform (DWT) and Wavelet Packet Transform (WPT) [3]. Although, REB vibration signal contains adequate fault data, they are non-linear and non-stationary in nature. Hence, extraction of signal feature is crucial for effective fault diagnosis. Several signal processing methods in time, frequency and Time- frequency domain like measuring the Root Mean Square (RMS), Peak, Standard deviation (SD), crest factor, Kurtosis(K), skewness, and shape factor have been proposed and implemented for effective bearing fault diagnosis [5–6]. The next procedure to be carried out in signal analysis is feature extraction and diagnosis of the fault. Out of time, frequency, and time–frequency domains, time–frequency method is productive in extracting features since non-stationary characteristics are exhibited by the bearing signals [3,6].

In the past, some authors [4] have presented a procedure for fault diagnosis of rolling element bearings through artificial neural network (ANN). Some other researchers [7], have proposed a methodology of fault feature extraction for roller bearings which is based on the intrinsic mode functions (IMFs) envelope spectrum. In addition, some other researchers [8], demonstrated that the feature vectors obtained by techniques such as Fast Fourier Transform (FFT), wavelet transform and bi-spectrum can be used as fault features and the Hidden Markov Model (HMM) as the classifiers to

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recognize the faults of the speed-up and speed-down process in rotating machinery. Various artificial intelligence techniques are used with wavelet transforms for fault detection in rotating machines [9–11]. A comparative experimental study for the effectiveness of ANN and support vector machine (SVM) in fault diagnosis of ball bearings was carried out and they concluded that the classification accuracy for SVM is better than ANN [12]. Some researchers [13], compared three sets of features and concluded that the frequency set of features yielded the best results. The accuracies of only frequency domain features (in comparison to the other two sets), increased when both SVM and KNN were applied. These authors [13] also showed that KNN classification results were better when compared to SVM using frequency domain features. DWT and CWT are compared for multi-class fault detection of an induction motor. However, Greedy-CWT method was used as it gave optimum solution for real-time motor fault diagnosis. The performance of the method did not hinder even in the presence of noise and low sampling frequency [14]. Researchers [15], have also used an enhanced KNN classifier to improve fault diagnosis of bearing. This resulted in significant improvement in diagnostic performance and the results were more robust to variations in the neighborhood size (k). In other studies [16], fault diagnosis was performed by fusing the data obtained from accelerometers located at different positions in the experimental setup. Simple statistical features such as elements of the Covariance matrix is used to identify the faults. The extracted features were given as input to K-NN which provided good results for Case Western Reserve University (CWRU) bearing data set. However, the scheme is to be verified with the experimental data. Some authors [17]. also proposed real-time online fault diagnosis approach for REB using KNN algorithm. They proved that real-time vibration diagnosis method using KNN yielded highest classification accuracy when compared to c4.5 algorithm and Classification and Regression tree (CART) algorithm. Authors[18], have also proposed the Particle swarm optimization (PSO) optimized KNN model for bearing fault diagnosis. The vibration signals were decomposed by local mean decomposition method and the product function are obtained from the decomposed signal. Then entropy values of the product functions were calculated by Shannon method, which were given as input to the KNN model. Kernel Principal Component analysis (KPCA) was used to reduce the data size and then PSO optimized KNN model is used to classify bearing faults. The proposed method is validated using the real word vibration signal. The result validated the effectiveness of the proposed method. Due to the presence of mechanical components, the vibration signals suffered from low signal-to noise ratio due to the surrounding environment and due to the existence of high frequencies, this will necessitate the determination of bearing defect frequencies in advance [6,13]. Many research outcomes justify the ML methods in machinery fault diagnosis methods as they are more useful than signal based methods [5–6,13]. Also, classification accuracy criterion of the fault diagnosis is aiding in selecting fault diagnosis approaches. In this paper, vibration signal pertaining to three conditions at different loads (0 and 1.7 kN) and maximum shaft speed (622 rpm) is acquired using a customized bearing test rig. Seventeen statistical features are extracted from the CWT coefficients. Extracted features were given as inputs to KNN classifier to classify the bearing conditions. The article has been divided into the following sections: section 2 provides description about CWT, section 3 and 4 explains the experimental setup and feature extraction, KNN classifier in section 5, section 6 presents results and discussions followed by conclusions in section 7.

2. Continuous wavelet transform

The main two types of WTs are CWT and DWT. The difference between the two can be observed in the way they discretize the scale parameter. CWT uses an exponential scale with a base smaller than 2, whereas the DWT uses an exponential scale with a base equal to 2. Hence, CWT discretizes the scale better than the DWT. CWT is more suitable for localizing transients in nonstationary signals. CWT has the same time resolution as the original data in each frequency band [5]. CWT is the best time–frequency transform for the analysis of nonstationary signals as reported in [5].

CWT indicates the correlation between the analyzing signal *f(t)* and the wavelet function which is stated by the equation (1). The CWT of *f(t)* is:

$$CWT(a,b) = \int_{-\infty}^{\infty} f(t)\Psi_{a,b}(t)dt \tag{1}$$

$$\Psi_{a,b}(t) = \frac{1}{\sqrt{a}}\Psi\left(\frac{t-b}{a}\right)$$

The signals are decomposed into functions called as wavelets, which are obtained from single basic wavelet called the mother wavelet. Here, $\Psi_{a,b}(t)$ denotes the mother wavelet. Parameter '*a*' represents scale index which in turn is the reciprocal of frequency and parameter '*b*' indicates time shifting. Continuously varying the values of scale parameter '*a*' and position parameter '*b*', CWT coefficients are obtained. Each coefficient multiplied with the corresponding scaled and shifted wavelet gives the constituent wavelets of original signal.

3. Experimental setup

A customized bearing test rig was used to acquire the vibration signals from REB. It consists of a shaft with 32 mm diameter supported between two bearings. The test rig setup is as shown in the Fig. 1. The test rig was arranged in such a way that it allows the shaft to be run at various speeds under variable radial loads. The shaft is driven by an induction motor with variable speed controller. Timer belt and pulley arrangement drives the motor and bearing test rig. For this setup, the speed ratio was 2.25. At one end of the shaft, a 6205 deep groove ball bearing is mounted on which radial loads are applied through hydraulic loading arrangement. Motor is made to run up to a speed of 1400 rpm. Table 1 shows the description of the test rig.

Vibration signals were acquired by placing two accelerometers on the housing of test bearing. After allowing the test rig to run for some time (for 5 s), vibration signals were acquired from the accelerometer mounted in X and Y direction. These signals were then fed to the Data Acquisition system installed in the computer through cables. The signals are collected at a sampling rate of

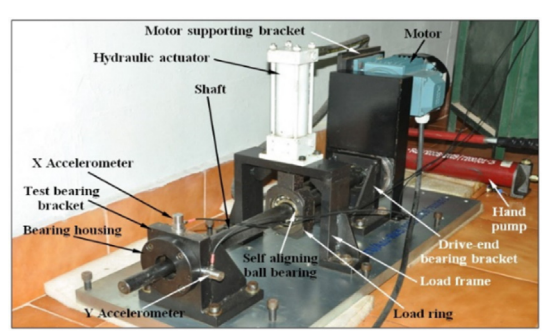

Fig. 1. Photographic view of REB test rig [21].

Table 1
Description of Experimental Setup and Data Acquisition System (DAQ).

Sl. No.	Item	Description
1.	Shaft diameter	32 mm
2.	Drive motor	3 phase, 1 HP induction motor
3.	Speed ratio	2.25
4.	Test bearing	6205 deep groove ball bearing
5.	Type of load applied	Two radial loads (0 and 1.7 kN) at 622 rpm
6.	Method of load application	Using a hydraulic actuator, load frame and a load ring
7.	Software used for data collection	LAB VIEW
8.	Software used for feature extraction	MATLAB-WAVELET TOOLBOX
9.	Software used for Statistical tests	IBM SPSS V20
10.	Sampling rate and time	48,000 samples/ second for 5.209 s
11.	Four independent conditions for which the data was acquired, and the symbols used	N – Healthy bearing.IR – Defective Inner Race. OR – Defective Outer Race
12.	Three pairs of bearing conditions created for statistical tests	N v/s. B, N v/s. IR and N v/s. OR

48,000 samples/sec. The signals were collected at specified sampling rate by a modified LAB VIEW software and were stored in computer as a.txt file.

Only signals from accelerometer in Y-direction were considered, as X-direction signals were found insensitive to bearing condition. CWT has been used to decompose the signal. A data set of 250000 × 1 resulted for each trial of the experiment. For two load, one speed and three conditions, six data vectors are obtained (two for each bearing condition). Fig. 2 shows the methodology used in this work.

4. Feature extraction

Vibration signals were acquired from bearing test rig for three conditions, i.e., N, IR and OR. Signals have been divided into 25 non-overlapping bins with each bin having 10,000 data. CWT is applied to the raw vibration signal with 1:128 scale with 'db8′ mother wavelet. Feature extraction is carried out using CWT coefficients. Thus, from the CWT coefficient matrix, eight features (listed in Table 2) are extracted which resulted in data size, 6 × 25. For 1.7 kN and no load and three conditions of REB, three such vectors (three for each load) are obtained. Thus, resulting in a matrix with dimension 8 × 150. The feature set is normalized so that the values lie in between 0 and 1. The feature set is randomly shuffled and partitioned with 80% of features is taken for training and remaining 20% features for testing of classifier.

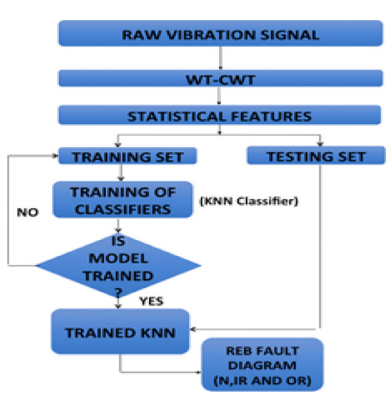

Fig. 2. Proposed Methodology for REB fault diagnosis.

5. KNN classifier

The nearest neighbor (NN) is a non-parametric technique which finds application in pattern recognition, text classification or ranking the models. One of the widely used algorithms in machine learning application is KNN as it is simple and good results were obtained when it is applied to diverse problems. KNN works by using an input vector with the K closest training samples in feature space. The algorithm requires training to define the neighbors based on the distance from the test samples and a testing step to determine the class to which this test samples belongs [17–18].

In KNN methodology, the idea is to allocate new unclassified data points to the class where majority of K nearest neighbors belong [19–20]. This algorithm is proven to be very effective in reducing the misclassification error even when the number of samples in dataset is large. The classification accuracy largely depends on the value of K where it represents number of neighbors. The decision rule of KNN can be written as:

$$f(x, C_j) = \sum_{ti \in kNN} sim(x, t_i) Z(t_i, C_j) \tag{2}$$

where × is the test data, t_i is the training data and sim(x, t_i) gives the similarity between them. Here C_j is the candidate category [19–20].

To carry out KNN algorithm, the steps are as follows:

1. The value of K is chosen based on the results obtained after some trails.
2. Distance measurement techniques like Euclidean and Manhattan are used. Distances obtained are sorted in ascending order and least k distance is chosen. The Euclidean distance between two points, say, P and Q is given by:

$$d(P, Q) = \sum_{i=1}^{n} (P_i - Q_i)^2$$

Classes of K nearest data are identified and the maximum ratio formed by this identified class is taken. This helps in providing the class of desired data.

6. Results and discussions

The objective of this work was to classify different conditions of the REB using KNN classifier. The KNN classification was implemented by classification learner app using MATLAB wavelet toolbox [22]. A scatter plot shows the data before training the classifier, and the model prediction results once the classifier has been trained. Fig. 3 shows scatter plot of the trained model (120 samples) with the classification accuracy of **91.79**% with **5**-fold cross validation for bearing condition classification using 17 TFD features. The classification accuracy of the KNN is explained using a confusion matrix shown in Table 3. Resultant matrix obtained is often called as the Confusion matrix and its diagonal elements represent properly classified data. Other than the diagonal elements, rest all represents misclassified data, and this can be attributed to the sensitivity of the feature with the defects. The terms used in the confusion matrix are A (Normal), B(Inner race) and C (Outer race). 80% of feature set, i.e., 120 instances were given for training set which resulted in a 91.79% accuracy for the training data as shown in Fig. 3. The remaining 20% were used for testing the trained KNN model. 25 out of 30 instances were properly classified with 3 in OR and 2 in IR being misclassified as IR and OR respec-

Table 2
List of statistical features.

Sl. No.	Features	Formula	Sl. No.	Features	Formula
1	Mean	$\frac{1}{n}\sum_{i=1}^{n} x_i$	10	Latitude Factor	P_k/SmSq
2	RMS	$\sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2}$	11	Shape Factor	Rms/Ma
3	Standard deviation	$\sqrt{\frac{1}{(n-1)}\sum_{i=1}^{n}(x_i - Mn)^2}$	12	Impulse Factor	P_k/Ma
4	Peak	$\frac{1}{2}[Max(x_i) - Min(x_i)]$	13	Standard deviation Impulse Factor	Sd/Ma
5	Square of Mean of Square root	$\left(\frac{1}{n}\sum_{i=1}^{n}\sqrt{\lvert x_i\rvert}\right)^2$	14	Log-log Ratio	$\frac{1}{\log(Sd)}\sum_{i=1}^{n}\log(\lvert x_i\rvert + 1)$
6	Mean of absolute value	$\frac{1}{n}\sum_{i=1}^{n}\lvert x_i\rvert$	15	Square root of ratio of square of mean of square root to standard deviation	$\sqrt{\frac{SmSq}{Sd}}$
7	Kurtosis	$\frac{1}{(n-1)}\sum_{i=1}^{n}\left(\frac{x_i - Mn}{Sd}\right)^4$	16	Normal Negative Log Likelihood	$-\sum_{i=1}^{n}\log\left[\frac{\exp\left\{\frac{-(x_i-Mn)^2}{2(Sd)^2}\right\}}{Sd\sqrt{2\pi}}\right]$
8	Crest Factor	P_k / Rms	17	Weibull Negative Log Likelihood	$-\sum_{i=1}^{n}\log\left[\beta\eta^{-\beta}\lvert x_i\rvert^{\beta-1}\exp\left\{\frac{\lvert x_i\rvert}{\eta}\right\}^{\beta}\right]$ where, β is the shape factor and η is the scale factor
9	Skewness	$\frac{1}{(n-1)}\sum_{i=1}^{n}\left(\frac{x_i - Mn}{Sd}\right)^3$			

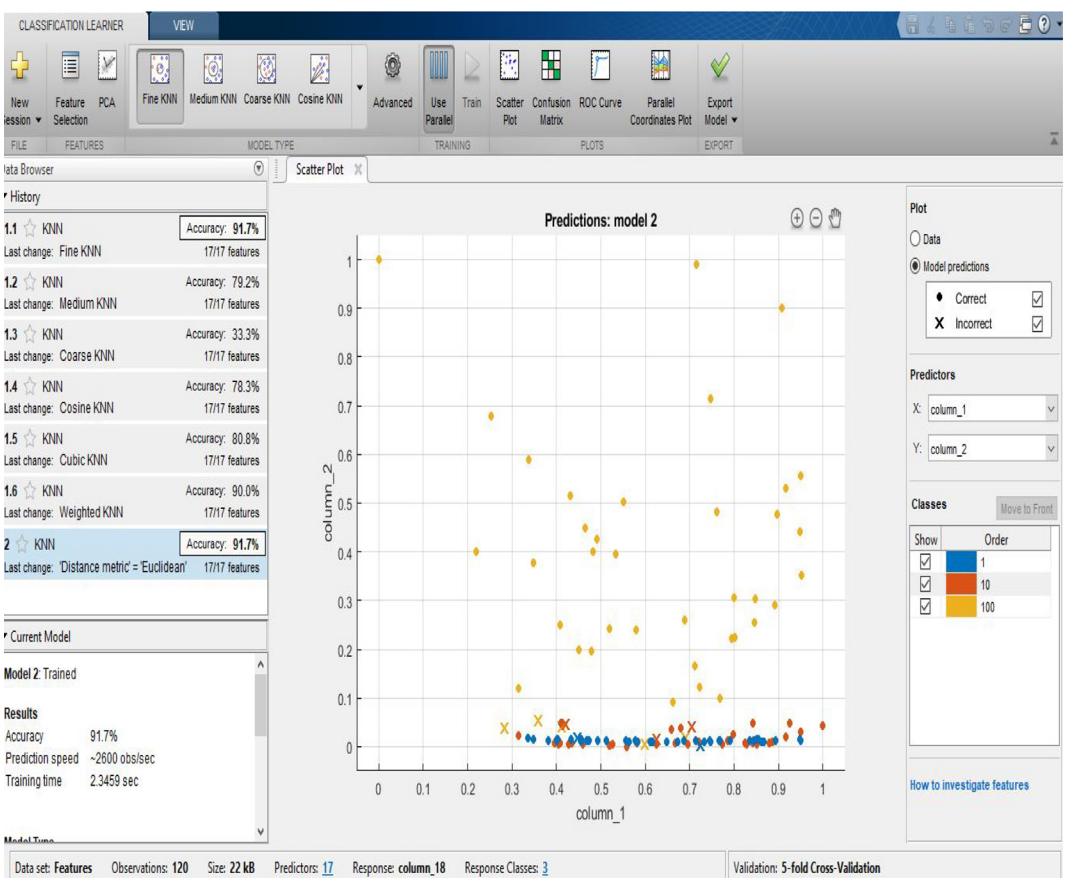

Fig. 3. Scatter plot of the trained KNN model.

Table 3
Confusion matrix of the KNN classifier testing sample for K = 1.

Confusion Matrix				Classification accuracy on test data (%)	Overall Classification accuracy on test data (%)
Actual class	Predicted class				
	A	B	C		
A	10	0	0	100	83.3
B	0	8	2	80	
C	0	3	7	70	

tively. This resulted in an overall accuracy of 83.33% for testing data. Thus, CWT in conjunction with KNN yielded satisfactory result and this method can be used for predicting the different conditions of REBs.

7. Conclusions

This study, employed a machine learning technique, revealed that the TFD features can be effectively used to classify different conditions of REBs. CWT has been applied to the raw vibration signal acquired at different load and maximum speed (622 rpm). Seventeen statistical features were extracted from the CWT coefficients. KNN classifier is trained using the classification learner app in MATLAB and the trained model is tested which resulted in classification accuracy of 83.3%. The test accuracy can be improved using the denoised signal and by employing other statistical features. Thus, the proposed method can be effectively used for the CM of REB.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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